Correcting AIFS Tropical Cyclone Intensity Forecasts: a Scaled-Down Reproduction of AIFS-TC

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September 2026

Abstract

AI weather models such as ECMWF's AIFS-Single forecast tropical cyclone tracks well but systematically underestimate intensity, because a 0.25° grid cannot resolve the eyewall. Allen et al. (2026) showed that a simple machine-learned correction, trained on tabular features of the forecast itself, removes most of this bias. This report reproduces the gradient-boosted half of that work at reduced scale, using only public data: 399 operational AIFS-Single forecasts covering every North Atlantic and East Pacific storm from February 2025 to September 2026, tracked with a purpose-built MSLP-minimum tracker and corrected with gradient-boosted trees under storm-grouped cross-validation. The raw model's maximum-wind mean absolute error (MAE) of 30.3 kt is more than halved to 14.4 kt, against 9.4 kt for the official NHC forecast. Raw-model error and the rapid-intensification failure mode match the paper almost exactly (71.4 kt vs. their 71.1 kt raw MAE on RI cases), while the corrected error remains above the paper's 11.0 kt, consistent with training on one and a half seasons instead of nine years. The tracker is validated against ECMWF's own operational AIFS tracks to within 1.1 kt MAE.

1 Introduction

Data-driven weather models have reached operational skill for synoptic-scale prediction, and ECMWF's AIFS-Single (Lang et al., 2024) runs operationally alongside the physics-based IFS. For tropical cyclones these models inherit a structural weakness: with grid spacing of about 28 km, the inner core of a hurricane is unresolved, so forecast 10 m winds saturate far below observed intensity even when the track is accurate. Allen et al. (2026) ("AIFS-TC") demonstrated that this bias is largely *predictable*: a gradient-boosted model over simple features of the AIFS forecast and the operationally known initial intensity brings AIFS intensity forecasts to parity with the operational frontier.

This project asks how much of that result survives a drastic reduction in resources: public data only, one and a half hurricane seasons instead of nine years of hindcasts, a home-made tracker, and a laptop. Beyond the replication itself, the by-product is an end-to-end open pipeline (catalog, fetch, track, correct, evaluate) of roughly 900 lines of Python.

2 Data

AIFS forecasts. The ECMWF Open Data AWS mirror (`s3://ecmwf-forecasts`, CC-BY-4.0) retains every operational AIFS-Single run since 2025-02-25. Storm lifetimes from the IBTrACS best-track archive (Knapp et al., 2010) define 399 forecast initializations (00/12 UTC) covering

all North Atlantic (NA) and East Pacific (EP) storms of at least tropical-storm strength through 2026-09-08. For each run, only the GRIB messages for mean sea-level pressure and 10 m winds are downloaded via HTTP byte ranges, cropped to 0–60°N, 180°W–0°, at leads 0–168 h every 6 h (7.1 GB total, versus hundreds of GB for the full archive).

Reduction caveats. The stored arrays are a lossy reduction: three surface variables, one basin pair, leads capped at 168 h (the mirror holds 360 h), and winds stored in half precision (error well below 0.1 kt, negligible against forecast error).

Operational archive, not hindcasts. Unlike the paper's controlled hindcast training set, these are the forecasts actually issued in real time. Two consequences matter. First, the model version changes mid-dataset: AIFS-Single 1.0 was operational for the 2025 season and 2.0 (released 2026-05-12) for 2026; a version flag is passed to the correction model and results are reported by season. Second, each run starts from the real-time operational analysis with its data-assimilation cutoff, not from a reanalysis.

Verification and benchmarks. Ground truth is the IBTrACS/ATCF best track (USA agency values). The benchmark is OFCL, the official human forecast of the National Hurricane Center, parsed from the ATCF a-deck archive, which also supplies GDMN, the ensemble mean of Google DeepMind's cyclone model (Google DeepMind, 2025), and EAIO, ECMWF's own tracker applied to the same AIFS runs.

3 Methods

Tracker. Each forecast is seeded at the best-track position at initialization. At each 6 h step the storm is located at the MSLP minimum within 450 km of a first guess (previous position advected with previous motion) on a lightly smoothed field; intensity is that minimum pressure and the maximum 10 m wind within 250 km. Tracking stops when the low is no longer distinct. Against ECMWF's operational AIFS tracks (EAIO) at matched initialization and lead ($n = 1209$, 24 storms), the tracker agrees to a mean difference of +0.8 kt and an MAE of 1.1 kt, so tracker choice is not a meaningful error source.

Correction model. Following the GBM branch of AIFS-TC, gradient-boosted trees (scikit-learn `HistGradientBoostingRegressor`, 5-seed average) predict the residual between best-track and AIFS maximum wind. The 22 features include the AIFS forecast state and its trends and running extremes, translation speed, basin, day of year, the model-version flag, storm age, the observed intensity and 12/24 h intensity trend at initialization, and the initial AIFS bias. Observed rapid-intensification cases (RI, ≥ 30 kt observed gain in 24 h) are weighted 2× in training, as in the paper. Evaluation uses storm-grouped 5-fold cross-validation, so a storm is never corrected by a model that saw any part of it. A persistence-of-initial-bias baseline (carrying the initialization-time AIFS bias forward unchanged) is included as a sanity check; the GBM beats it at every lead beyond 24 h.

4 Results

Table 1 and Figure 1 give the headline result on the homogeneous sample (cases where raw, corrected, and OFCL forecasts all exist and verify against best track): the correction cuts overall MAE

Table 1: Maximum-wind MAE (kt) versus best track by forecast lead time, homogeneous sample ($n = 2223$ cases, 46 storms, NA+EP, Feb 2025–Sep 2026). Bottom row: all leads pooled, with 95% storm-bootstrap confidence intervals.

Lead (h)	n	AIFS raw	AIFS+GBM	NHC OFCL
12	442	26.4	10.8	4.9
24	396	28.4	13.2	7.4
36	358	29.8	14.9	9.5
48	315	31.4	16.3	10.9
72	247	33.4	16.6	11.5
96	190	35.2	16.7	12.8
120	148	35.8	16.4	14.1
144	74	31.7	16.3	13.6
168	53	25.0	14.2	11.1
12–168	2223	30.3 [24.5, 35.1]	14.4 [12.4, 16.6]	9.4 [8.0, 10.5]

Table 2: Pooled (12–168 h) maximum-wind MAE (kt) with 95% storm-bootstrap confidence intervals, by subset.

Subset	n	Storms	AIFS raw	AIFS+GBM	NHC OFCL
All cases	2223	46	30.3 [24.5, 35.1]	14.4 [12.4, 16.6]	9.4 [8.0, 10.5]
RI cases only	162	13	71.4 [61.7, 77.5]	38.3 [32.0, 42.8]	20.2 [16.5, 23.5]
North Atlantic	772	15	32.4 [22.0, 39.3]	15.5 [10.7, 19.7]	10.4 [7.7, 12.4]
East Pacific	1451	31	29.2 [22.2, 35.7]	13.9 [11.4, 16.2]	8.9 [7.4, 10.2]
2025 (AIFS 1.0)	1526	30	29.6 [23.2, 35.5]	15.5 [12.9, 18.1]	9.6 [8.2, 10.9]
2026 (AIFS 2.0)	697	16	31.8 [20.1, 39.2]	12.1 [8.4, 14.5]	8.8 [5.8, 10.9]

from 30.3 to 14.4 kt, with non-overlapping 95% confidence intervals from a storm-level bootstrap. It does not reach OFCL (9.4 kt) overall, but approaches it at long leads (16.4 vs. 14.1 kt at 120 h). On a separate homogeneous sample containing GDMN ($n = 2022$, 41 storms), DeepMind's dedicated cyclone model scores 9.9 kt, on par with OFCL (9.6), with corrected AIFS at 14.6.

Table 2 breaks the pooled result down. The correction helps in both basins and both seasons. Notably, it performs better on the 2026 season (AIFS-Single 2.0, 12.1 kt) than on 2025 (v1.0, 15.5 kt) despite 2026 contributing less training data, suggesting the newer model's errors are more systematic and therefore easier to learn. Rapid intensification remains the hard case: raw AIFS misses RI by 71.4 kt on average, the correction recovers roughly half of that (38.3 kt), and the human forecast (20.2 kt) stays well ahead.

Figure 2 shows the raw bias at its most dramatic: Hurricane Lowell (EP, 2026) reached 140 kt while every raw AIFS forecast saturated near 60 kt. Figure 3 shows the underlying fields and tracks for one initialization with three simultaneous storms.

5 Comparison with AIFS-TC

Table 3 sets these results against the paper's NA+EP evaluation (Allen et al., 2026). The samples differ (their held-out 2025 under one model version and hindcast initial conditions; here 2025–2026 operational output under two versions), so each column should be read against its own OFCL, not

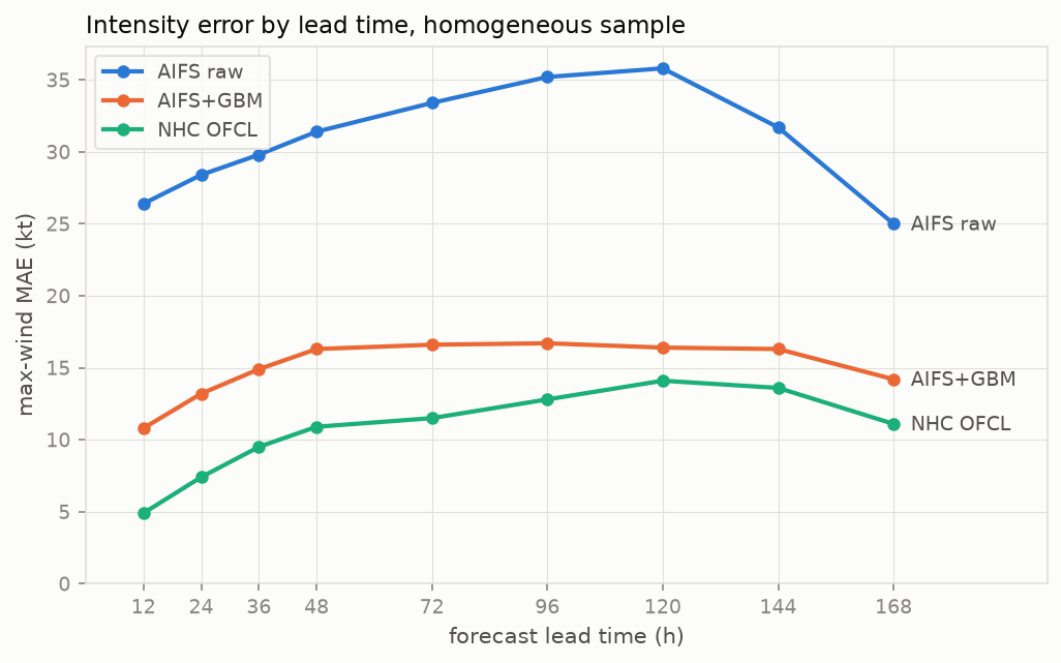


Figure 1: Maximum-wind MAE by lead time on the homogeneous sample. The raw AIFS error is dominated by the under-resolution bias; the GBM correction removes most of it at every lead.

Table 3: Pooled maximum-wind MAE (kt), this work versus AIFS-TC (their NA+EP test set, 29 storms). Samples differ; compare each system to its own sample's OFCL.

	This work	AIFS-TC
AIFS raw	30.3	35.3
Corrected	14.4	11.0
NHC OFCL	9.4	10.7
AIFS raw, RI only	71.4	71.1
Corrected, RI only	38.3	20.9
OFCL, RI only	20.2	20.1

across columns.

Three observations. First, the *baselines* land on the paper's numbers: raw AIFS error is the same phenomenon at the same magnitude, and on RI cases the agreement is striking (71.4 vs. 71.1 kt). Together with the 1.1 kt tracker agreement, this indicates the pipeline measures the same quantity the paper measures. Second, the *correction* reproduces the qualitative claim, recovering roughly three quarters of the raw-to-OFCL gap, but not the paper's parity with OFCL. Third, the shortfall concentrates exactly where limited training data should hurt most: rare RI events, of which this dataset contains 13 storms' worth against the paper's nine years. The 2026-only result (12.1 kt, single model version) already approaches the paper's 11.0.

6 Limitations

One and a half seasons versus nine years of training data; two model versions in one dataset, handled only by a feature flag; the GBM branch only (no CNN over three-dimensional patches);

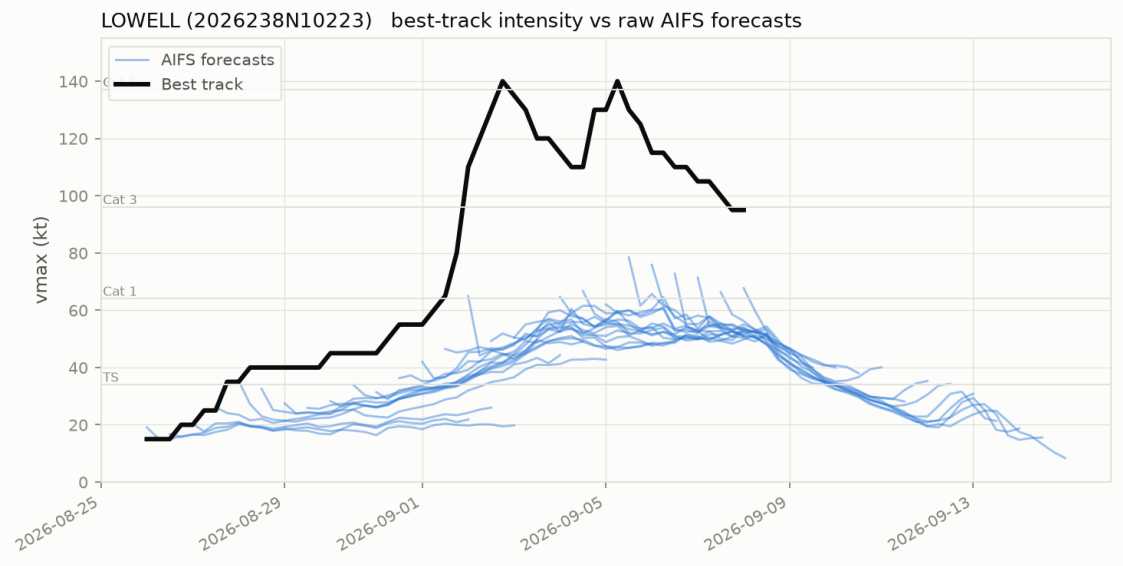


Figure 2: Observed intensity of Hurricane Lowell (2026, EP; black) against all 27 raw AIFS forecasts (blue). The model's track is good but its intensity saturates near 60 kt against an observed peak of 140 kt: the under-resolution bias the correction targets.

best-track initial intensity as a feature where fully operational practice would use the real-time CARQ estimate (the paper reports both variants); two basins; leads capped at 168 h. Position errors are measured but not corrected.

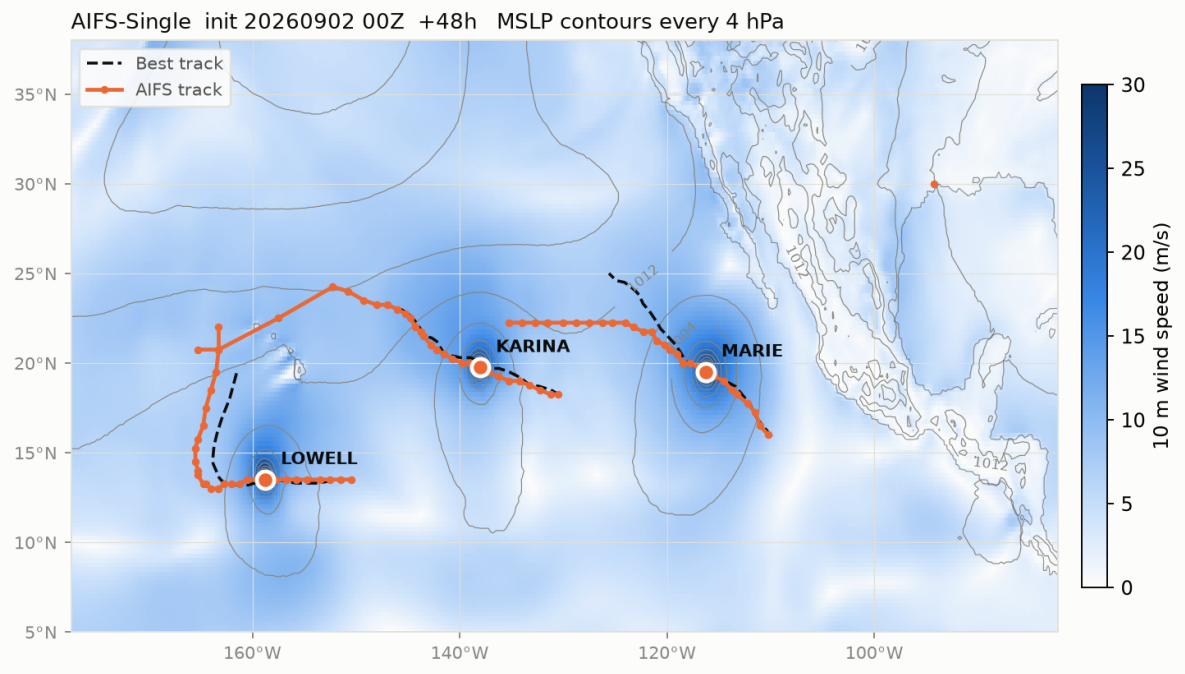


Figure 3: AIFS-Single forecast fields at +48 h from the 2026-09-02 00Z run: 10 m wind speed (shading), MSLP (contours every 4 hPa), forecast tracks (orange) and best tracks (dashed) for hurricanes Lowell, Karina and Marie.

7 Reproducibility

All code is at `https://github.com/MichelFaloughi/scmp-5990` (directory `tc/`). The 7.1 GB of forecast data regenerates from public sources with two commands (catalog and fetch); the full pipeline runs on a laptop in minutes thereafter. Data sources: ECMWF Open Data (CC-BY-4.0), IBTrACS v04r01, and the NHC ATCF archive.

References

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